Determine where , if anywhere , the function $y = 2z^4 - z^3 - 3z^2$ *is not changing .*

$$\frac{dy}{dz} = 8z^3 - 3z^2 - 6z$$

$$\frac{dy}{dz} = 0$$

$$\rightarrow z(8z^2 - 3z - 6) = 0$$

$$z = 0$$

$$8z^2 - 4z - 6 = 0$$

$$z = \frac{1}{4} + \frac{\sqrt{13}}{4}$$

$$z = \frac{1}{4} - \frac{\sqrt{13}}{4}$$